

Heaven's Light Is Our Guide
RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
1st Year EVEN Semester Examination 2024
COURSE NO: CSE 1201 COURSE TITLE: Data Structure
FULL MARKS: 60 TIME: 3 HRS

- N.B. (i) Answer any SIX questions taking any THREE from each section.
(ii) Figures in the right margin indicate full marks and Course Outcomes (COs).
(iii) Use separate answer script for each section.

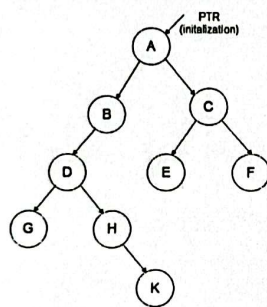
Marks COs

SECTION : A

Q.1.	(a)	Define different data structure with at least one application for each of them.	04	CO1																																	
	(b)	Explain why it is needed to study data structure in terms of complexity.	03	CO2																																	
	(c)	Suppose you have the following data. Explain any three data structure operations on the following data. 100, 12, 20, 200, 50, 80	03	CO3																																	
Q.2.	(a)	Suppose you are designing a marks processing system for two different classes: – ClassA[] array contains marks of students in ascending order – ClassB[] array contains marks of students on ascending order i) Describe the array operations needed to insert a new student's mark into ClassA[] without breaking the sorted order ii) Draw a flowchart to merge ClassA[] and ClassB[] into AllMarks[] in ascending order. iii) Write an algorithm to split AllMarks[] into two arrays: PassList[] (marks ≥ 40) and FailList[] (marks < 40)	10	CO2																																	
Q.3.	(a)	Write the advantages and disadvantages of using linked list over array data structure.	03	CO1																																	
	(b)	Explain the complexity of the following function in big O notation. <pre>void f(n){ int i,j,k,l; for(i=1;i≤n;i++) for(j=0;j≤i;j++) for(k=1;k≤j;k++) for(l=1;l≤100;l++) printf("CSE 24\n"); }</pre>	03	CO2																																	
	(c)	Suppose you have a singly linked list as depicted below. Apply linked list searching algorithm to find the data=5 (at least after 4/5 operations). START 0 ↓ <table border="1" style="border-collapse: collapse; text-align: center; width: 100%;"><tr><td></td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>info</td><td>1</td><td>6</td><td>2</td><td>7</td><td>3</td><td>8</td><td>4</td><td>9</td><td>5</td><td>10</td></tr><tr><td>link</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>		0	1	2	3	4	5	6	7	8	9	info	1	6	2	7	3	8	4	9	5	10	link											04	CO3
	0	1	2	3	4	5	6	7	8	9																											
info	1	6	2	7	3	8	4	9	5	10																											
link																																					
Q.4.	(a)	Suppose you are developing a music playlist App where songs are connected in such a way that the user can move forward and backward between them. i) Identify the type of linked list suitable for this scenario and justify your choice. ii) Draw a flowchart to insert a song after a given song title. iii) Explain how deletion is handled when removing the last song in the list.	10	CO1																																	

SECTION : B

- Q.5. (a) Suppose you are designing a system to store student IDs in sorted order for fast searching. In the context, explain how a Binary Search Tree (BST) provides better performance than traditional binary tree. 03 CO4
- (b) Show how the BST changes after inserting these IDs: 45, 30, 70, 20, 40, 60, 80. 03 CO4
- (c) Explain the three cases of node deletion in a BST 04 CO4
i) leaf node, ii) node with one child and iii) node with two children
- Q.6. (a) Define i) STACK ii) Polish notation 02 CO1
- (b) Convert the following infix expression into its equivalent postfix expression: 04 CO2
i) $a(b+c)/d - e * (f+g)/h$ ii) $((a+b)/d) \wedge ((c-f) * g)$
- (c) Suppose you have the following postfix expression: 04 CO3
5, 6, 2, +, *, 12, 4, /, -
Do the followings: i) convert it into equivalent infix expression and evaluate value of it ii) Evaluate it using stack.
- Q.7. (a) Explain the concept of collision in hash tables and why it occurs. 03 CO3
- (b) Apply Heap sort on the array [4, 10, 3, 5, 1] and show the heap after each extraction. 03 CO3
- (c) Compare the any four sorting algorithms in terms of time complexity, space complexity and stability. 04 CO3
- Q.8. (a) Define i) Circular Queue and ii) Priority Queue 02 CO4
- (b) Find the sequence of nodes when the following tree are traversed by 05 CO2
i) inorder, ii) preorder and iii) postorder traversing using STACK.



STACK	
7	
6	
5	
4	
3	
2	
1	10
TOP	

- (c) Suppose you have the following sequence of numbers: 03 CO3
44 30 50 22 60 77 55
Now prepare a MAX heap.

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

1st Year Even Semester Examination 2023

COURSE NO: CSE1201

COURSE TITLE: Data Structure

FULL MARKS: 60

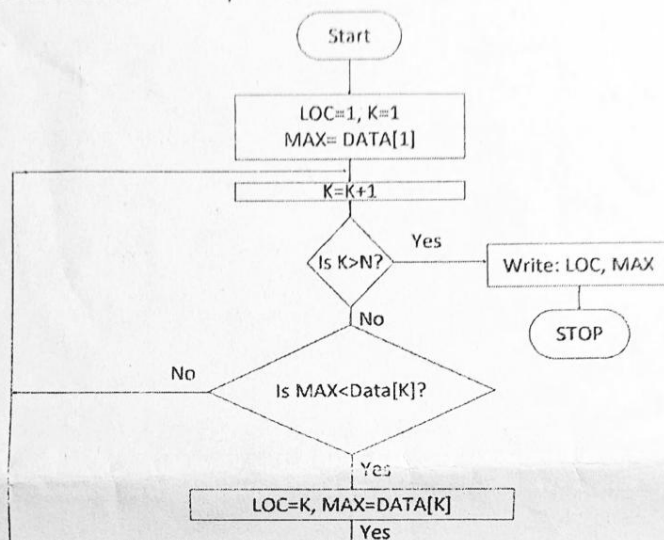
TIME: 3HRS

- N.B. (i) Answer any SIX questions taking any THREE from each section.
 (ii) Figures in the right margin indicate full marks.
 (iii) Use separate answer script for each section.

SECTION: A

Marks CO

- Q.1. (a) Define Data Structure. Briefly describe Data Structure operations. 5 CO1
 (b) Explain (i) Worst case (ii) Best case and (iii) Average case complexity of an algorithm. 3 CO3
 (c) Suppose you have the following flow chart in order to solve a particular problem.
 (i) Write steps to solve the problem.
 (ii) Write the name of the problem.



2 CO3

Here symbols represent as usual meanings in Data Structure.

- Q.2. (a) What is the difference between data and information? Give example for each of them. 2.5 CO1
 (b) Explain: (i) Primitive Data Structure
 (ii) Non-primitive Data Structure and
 (iii) Abstract Data Type <ADT> 4.5 CO2
 (c) Calculate the complexity of the following function in big O notation.

```

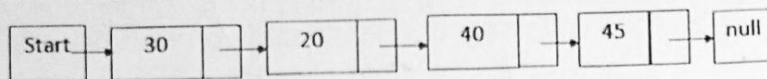
void A(n){
    inti, j, k;
    for(i=1; i<=n; i++)
        for(j=1; j<=i; j++)
            for(k=1; k<=100; k++)
                printf("CSE 22\n");
}
  
```

 3 CO3
- Q.3. (a) What is an Array Data Structure? Write advantages and disadvantages of using it. 3 CO1
 (b) Explain how can you sort the following data using Bubble Sort Algorithm. 32, 51, 27, 85, 66 3 CO2
 (c) Suppose an unsorted array of n numbers of integers has given to you. Write a complete program to find the length of the longest consecutive elements sequence. Show the sample input and output. 4 CO3
- Q.4. (a) Translate, by inspection and hand, each infix expression into its equivalent postfix expression. 6 CO3
 (i) $A*(b+c)/d-e*(f+g/h)$
 (ii) $((a+b)/d)\uparrow((c-f)*g)$
 (b) Based on Ackerman function, evaluate the following $A(2, 1)$. 4 CO3

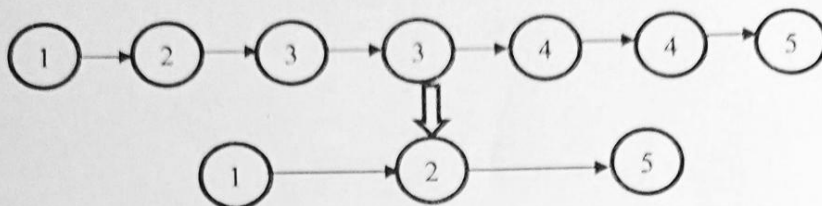
SECTION : B

- Q.5. (a) Briefly describe how does a linked list differ from an Array Data Structure? 2 CO1
 (b) Explain how to insert a new node at the end of an existing single linked 4 CO2

list. Also write a function to implement the above.



- (c) Suppose you have given the head of a sorted linked list as illustrated bellow figure. Write a code to delete duplicate numbers from it.



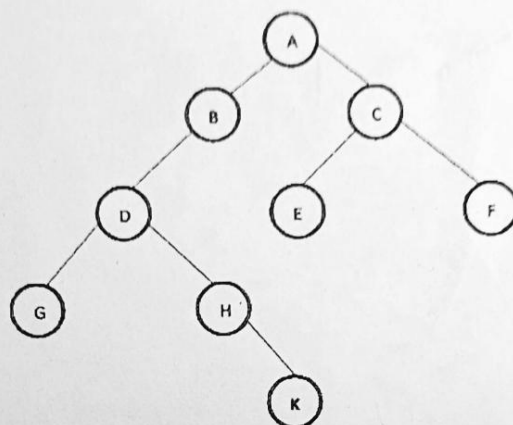
4 CO3

Input: head=[1, 2, 3, 3, 4, 4, 5]

Output: [1, 2, 5]

- Q.6. (a) Briefly describe the recursion for the Tower of Hanoi problem of 5 disks.
(b) Explain the preorder traversing using stack for the following tree.

2 CO1



4 CO2

- (c) Consider a ticketing system for a movie theater where customers arrive at different times and served in the order they arrive. Model this system using a queue to manage the customers waiting for their tickets. What changes would you make to your implementation if the queue had a maximum capacity and customers arrive after the queue is full?

4 CO3

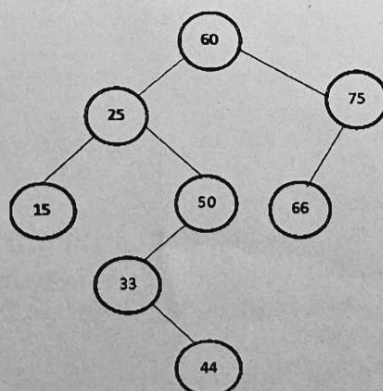
- Q.7. (a) Using linear probing, insert the following keys into a Hash table of size 10, with a Hash function $h(k) = k \bmod 10$: {12, 44, 13, 88, 23, 94, 11, 39, 20, 16}. Show the Hash table after all insertions, and indicate where collisions occur and how they are resolved?
(b) Given the following sequence of numbers, construct a BST and show the resulting tree after all the elements are inserted.
50, 30, 10, 20, 70, 40, 60, 90, 80.

6 CO4

4 CO4

- Q.8. (a) Discuss Hash technique and hence write the advantages and disadvantages of using it.
(b) Is the following Tree BST? If so, make it AVL tree using Balanced Factor (BF) calculation.

3 CO1



4 CO2

- (c) Suppose you have the following data: 100, 5, 25, 15, 10, 20. Apply (i) Selection and (ii) Insertion sort algorithm on this data and hence compare between them.

3 CO3

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
1st Year Even Semester Examination 2022
COURSE NO: CSE 1201 **COURSE TITLE: Data Structure**
FULL MARKS: 60 **TIME: 3 Hours**

- N.B. (i) Answer any **SIX** questions taking any **THREE** from each section.
(ii) Figures in the right margin indicate full marks.
(iii) Use separate answer script for each section.

Marks

SECTION A

- Q.1. (a) Define Data Structure and hence explain (i) primitive (ii) Non-Primitive (iii) Linear and (iv) Non-Linear Data Structure. [CO1] 05
(b) What is Sparse Matrix? Write the advantages of using it. [CO1] 03
(c) Consider the following domains of applications and mention at least one instance where sorting or searching can be utilized: (i) Databases (ii) E-Commerce (iii) File organization and (iv) Data Visualization. [CO1] 02
- Q.2. (a) Briefly describe the following Data Structure Operations: (i) Traversing (ii) Searching (iii) Inserting and (iv) Merging. [CO1] 04
(b) Explain how does the complexity of Algorithm so important? [CO2] 03
(c) Differentiate between a triangular matrix and a tri-diagonal matrix with suitable example. [CO1] 03
- Q.3. (a) With example how does a linear search algorithm follow (i) Best Case Time Complexity; $c(n) = 1$ (ii) Worst Case Time Complexity; $c(n) = n$ and (iii) Average Case Time Complexity; $c(n) = (n+1)/2$. [CO2] 05
(b) Suppose you have an array DATA of numerical values is in memory. Write algorithmic steps to find the location LOC and the value MAX of the largest element of DATA. [CO3] 02
(c) Imagine a music player application where songs are organized in a playlist as a Linked List, allowing users to skip forward or backward through songs. If the current song being played crashes the application due to a corrupt file, how might a Linked List be beneficial in resolving the issue? Express with proper illustration. [CO2] 03
- Q.4. (a) Apply the (i) Bubble Sort and (ii) Quick Sort Algorithm on the following DATA and hence compare the complexity. [CO2] 04
DATA: 77, 42, 35, 12, 101, 5
(b) Suppose you have the following DATA. Let's find the DATA 77 using Binary Search Algorithm. [CO3] 03
DATA: 10, 22, 25, 44, 55, 66, 77, 79, 92, 99
(c) Suppose you are designing an application for book navigation. You want to let the users move forward or backward while reading a book. Illustrate this phenomenon with the help of a two-way Linked List and a circular header Linked List. Which one is better as per your opinion? Justify. [CO2] 03

SECTION B

- Q.5. (a) Suppose you are training your computer to solve the elementary math. You want it to solve the following arithmetic expression: $5 * (6 + 2) - 12/4$. Which notation will you be using? Mention why your chosen notation will be helpful for computer while computation? [CO3] 04
(b) Explain and write an algorithm to merge two sorted Linked List pointed by h1 and h2 respectively into one sorted Linked List. Use h1 or h2 as the output List. Do not use the third Linked List. [CO2] 04
(c) Stack can be used as a mean of 'postponed Decisions'-Justify this statement. [CO3] 02
- Q.6. (a) Consider the following double ended queue (input restricted in front). It can be 04

utilized while booking trains. Now apply the following operations: (i) Push 'Ean' (ii) Push 'Jnan' (iii) Pop (iv) Pop (v) Pop (vi) Push 'Hatim'. [CO3]

- (b) How to move the disk for Tower of Hanoi problem of 4 disks. Suppose Ackerman function is defined as

$$A(m, n) = \begin{cases} n + 1; & \text{if } m = 0 \\ A(m - 1, 1); & \text{if } m > 0 \text{ and } n = 0 \\ A(m - 1, A(m, n - 1)); & \text{if } m > 0 \text{ and } n > 0. \end{cases}$$

Calculate $A(1, 2)$. [CO3]

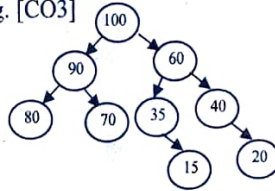
- (c) Write the difference between STACK and QUEUE Data Structure. [CO1]

- Q.7. (a) Differentiate between tree and binary search tree (BST). Consider the following data sequence and build a BST. [CO4]

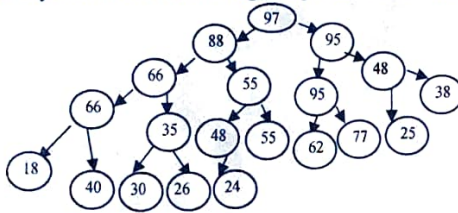
11, 6, 8, 19, 4, 10, 5, 17, 43, 49, 31

- (b) Reconsider the BST of Q.7. (a) and apply the following operations: (i) Delete 5 (ii) Delete 8 and (iii) Delete 19. [CO4]

- (c) Suppose you have the following Tree. Write the sequences using Pre-order, post-order and in-order traversing. [CO3]



- Q.8. (a) Suppose you have the following Heap. Insert 70 into this tree. [CO3]



- (b) There are the four types of notations in AVL trees. Apply them for the following data sequences where $x < y < z$. [CO4]

(i) x, y, z (ii) z, y, x (iii) x, z, y (iv) x, x, y .

- (c) Consider the following data sequence and build an AVL tree. [CO4]

14, 17, 11, 7, 53, 4, 13, 12, 8, 60, 19, 16, 20

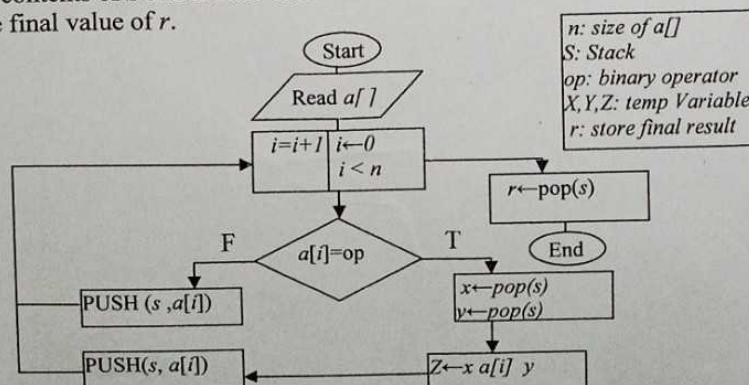
Heaven's Light Is Our Guide
RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
1st Year Even Semester Examination 2021
COURSE NO: CSE 1201 COURSE TITLE: Data Structure
FULL MARKS: 60 TIME: 3 HRS

- N.B. (i) Answer any **SIX** questions taking any **THREE** from each section.
(ii) Figures in the right margin indicate full marks.
(iii) Use separate answer script for each section.

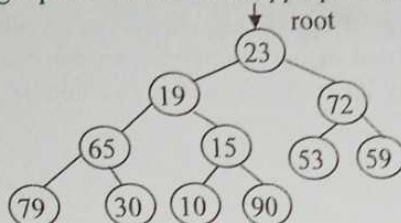
SECTION : A		COs	Marks
Q.1. (a)	Explain with example, i) Abstract Data type, ii) Linear Data type, iii) Non-Linear Data type, and iv) Time-space Trade-off.	CO1	4
(b)	The time complexity is a measure of a good algorithm. For an algorithm, the time complexity must be calculated before it applying to solve a real-world problem. Find the time complexity $T(n)$ of the following function $A()$	CO1	3
<pre> void A(n){ int i=0, s=0; while(s<=n){ i=i+1; s=s+i; cout<<s; } } </pre>			
(c)	A sparse matrix plays an important role specially when we transfer a large media file through computer network. A media file can be represented by a sparse matrix that contains many zero elements. To deal with a sparse matrix we need to store it first. Then draw a flowchart to store a 40×50 sparse matrix using a 2D array.	CO1	3
Q.2. (a)	Define and explain two way linked list data structure.	CO2	3
(b)	Design a linked list data structure such that the following operations can be occurred successfully. (i) Insertion of a new data (ii) Deletion of a data after (i)	CO2	5
(c)	How to represent a linked list in sequential memory?	CO2	2
Q.3. (a)	Write pseudo code statements to create one-node double circular linked list.	CO2	3
(b)	Suppose a single linked list headed by a pointer h . Then write an algorithm (flowchart) to delete all x -valued nodes. Take x from input.	CO2	4
(c)	Write a algorithm (flowchart) to convert a single linked list headed by h into a circular linked list.	CO2	3
Q.4. (a)	What is (i) Array (ii) Linked List data structure?	CO1	3
(b)	Develop an algorithm such that the following data is sorted quickly in comparison to others. 66, 32, 85, 51, 27, 20, 80, 90, 10	CO1	4
(c)	How can you measure the performance of Bubble Sort Algorithm?	CO1	3

SECTION : B

- Q.5. (a) In computability theory, the Ackermann function named after Wilhelm Ackermann. Find the value of Ackermann function (2,1). CO3 3
- (b) Convert the infix expression $(A+B-C) \times D / (E+F)$ to its i) Postfix and ii) prefix equivalent. CO3 3
- (c) Evaluate the value of postfix expression $a[] = "23+92-5+*2/"$ using the following algorithm. The numbers inside the expression are single-digit number. CO3 4
- Also find-
- i) the contents of S and Z after each value of i and
- ii) the final value of r .



- Q.6. (a) What is Queue data structure? Write the real life example of it. CO3 3
 (b) Design a Queue data structure such that it can be represented using Linked List. Explain with example. CO3 4
 (c) With examples, write the application of- i) Dequeue, and ii) Priority Queue. CO3 3
 Q.7. (a) Derive an expression of Time Complexity $T(n)$ for the following cases when Quick Sort algorithm is implemented to sort a set of n numbers- CO4 3
 i) Best case, ii) Average case, iii) Worst case.
 (b) Max-heapify the following binary tree. After that do the following CO4 3
 i) delete the root
 ii) after deleting replace the root with appropriate node and adjust the heap.



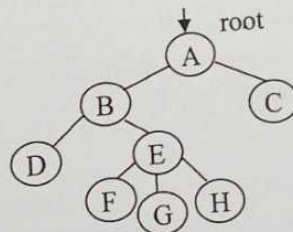
- (c) To sort an array $a[5]=\{3,2,8,7,5\}$ using Merge sort Technique, we need to do the partition first using a recursive method. Then draw a recursive calling sequence diagram for the following MergeSort() method with $low=0$ and $high=4$ initially- CO4 4

Pseudo code for MergeSort() method:

```

MergeSort(a, low, high){
    if(low < high){
        mid ← (low + high) / 2;
        MergeSort(a, low, mid);
        MergeSort(a, mid + 1, high);
    }
}
  
```

- Q.8. (a) If an array $a[] = \{10, 30, 35, 80, 45, 65, 75, 15, 20, 60, 85, 90, 95\}$ is stored as a binary search tree then find- CO4 3
 i) the right child of 80
 ii) the left child of 45
 iii) the parent of 90
 iv) the list of Leaf Nodes
 (b) Create a Binary Search Tree (BST) with linked list representation using the following node values- CO4 3
 50, 80, 40, 30, 45, 60, 75, 72, 42
 Now draw the above BST after-
 i) deleting node 42 (total tree nodes=8)
 ii) deleting node 75 (total tree nodes=8)
 iii) deleting node 40 (total tree nodes=8)
 (c) Obtain a Left-children/Right-Sibling representation table for the following general tree. CO4 4



*** END ***

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
1st Year Even Semester Examination 2019
COURSE NO: CSE 1201 COURSE TITLE: Data Structure
FULL MARKS: 72 TIME: 3 HRS

- N.B. (i) Answer any SIX questions taking any THREE from each section.
(ii) Figures in the right margin indicate full marks.
(iii) Use separate answer script for each section.

<u>SECTION : A</u>		Marks
Q.1.	(a) What is data structure? Describe three major operations playing vital role in data structure.	4
	(b) Explain time-space tradeoff with an appropriate example.	4
	(c) Draw a flow chart to insert an element ITEM into a linear array LA with N elements at K-th position.	4
Q.2.	(a) Suppose the following numbers are sorted in an array A: 45, 87, 15, 25, 19, 65, and 74. Describe every step to sort the array using- bubble sort algorithm and count the total number of comparisons required.	6
	(b) Let we have a (lower) triangular array A and we want to store nonzero entries of A in a linear array B as $B[L]=a_{JK}$, Find the relation among L, K and J.	3
	(c) Find the time complexity of binary search algorithm.	3
Q.3.	(a) What is meant by the "stack overflow" condition? Is it applicable to the linked list method of implementation of the stack? Give reason.	4
	(b) Design a data representation which sequentially map n objects into an array $a[1, n]$, $n1$ of these data objects are stacks and the remaining $n2$, equal to $n-n1$ are queue. Write an algorithm to add and delete elements from these objects.	6
	(c) The following sequence of operation is performed on a stack: push(1), push(2), pop, push(1), push(2), pop, pop, pop, push(2), pop. Determine the sequences of popped out values.	2
Q.4.	(a) Show, with the help of an example, how the limitations of an array can be avoided by "linked list". What do you mean by linear list and generalized list?	4
	(b) Let A and B be two linked lists. Write an algorithm to create a new linked list C that contains elements alternately from A and B beginning with the first element of A. If you run out of elements in one of the lists, then append the remaining elements of the other list to C.	5
	(c) What is circular linked list? When does circular linked list is beneficial?	3

SECTION : B

- | | | |
|------|---|---|
| Q.5. | (a) Draw a flow chart to insert an ITEM into a queue representing by a linear array. | 4 |
| | (b) Write down the steps to add an ITEM with priority number N to a priority queue which is maintained in memory as a one-way linked list. | 4 |
| | (c) Consider the following deque of numbers where DEQUE is a circular array which is allocated six memory cells:
LEFT=2, RIGHT=4, DEQUE: <u> </u> , 10, 23, 34, <u> </u> , <u> </u> . | 4 |

Describe the deque while the following operations take place:

- (i) 14 is added to the right of the deque.
- (ii) Two number on the left are deleted.
- (iii) 20 is added to the left of the deque.
- (iv) One number on the right is deleted.

- Q.6. (a) Consider the following binary search tree which contains maximum 13 nodes. Draw a linked representation which uses three parallel arrays. 4

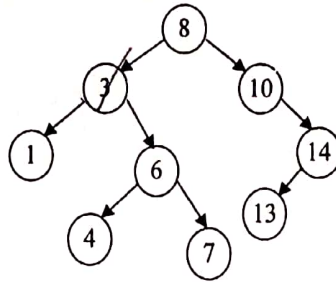


Fig 6(a)

- (b) Traverse this binary search tree according to pre-order, in-order and post-order technique. 4
- (c) Describe the each step of deletion a node from a binary search tree and delete the node of the above figure 6(a) which contains 3. 4

- Q.7. (a) Build a Huffman tree from the following frequencies table: 3

A	0.20
B	0.04
C	0.07
D	0.11
E	0.32
F	0.06
G	0.05
H	0.15

- (b) For the following binary tree answer the following: 5

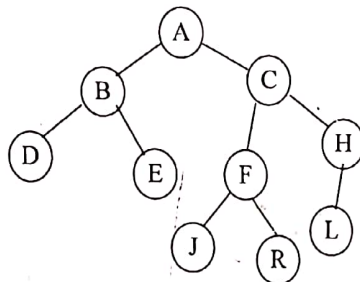


Fig 7(b)

- (i) What is the height of the tree?
- (ii) What are the ancestors and descendant of node C and H?
- (iii) What is the pre-order traversal?
- (iv) Degree of tree.
- (c) What is the difference between a heap and a binary search tree? 2
- (d) There are 8, 15, 13, 14 nodes in four different trees. Which of them could have formed a full binary tree? 2
- Q.8. (a) Show the different passes required to perform selection sort on the following set of numbers: 4
- 76, 32, 43, 10, 87, 21, 65, 54
- (b) What are the worst case and average complexity of the followings: 4
- (i) Insertion sort
- (ii) Selection sort
- (iii) Merge sort
- What are the benefits and limitations of merge sort?
- (c) What is meant by hashing and rehashing? How do you solve hash clashes by open addressing method. 4

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

1st Year Even Semester Examination 2018

COURSE NO: CSE 1201 COURSE TITLE: Data Structure

FULL MARKS: 70 TIME: 3 HRS

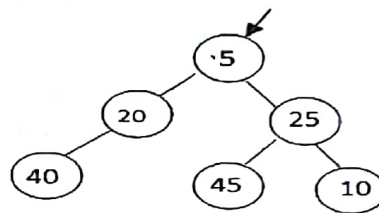
- N.B.** (i) Answer any **SIX** questions taking any **THREE** from each section.
(ii) Figures in the right margin indicate full marks.
(iii) Use separate answer script for each section.

SECTION : A

- Q.1. (a) What do you mean by data structure? Write the advantages of studying data structure course. 4
(b) Briefly describe data structure operations. $3 \frac{1}{2}$
(c) Explain (i) worst case (ii) average case and (iii) best case complexity of linear search algorithm for n input data. $4 \frac{1}{2}$
- Q.2. (a) What is sparse matrix? 2
✓ (b) Convert the following sparse matrix into triplet representation. 3
- | | | | | | |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 9 |
| 0 | 8 | 0 | 0 | 0 | 2 |
| 1 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 5 | 0 |
| 0 | 0 | 4 | 0 | 0 | 0 |
- (c) Discuss the limitations of array processing. 2
(d) Let S1 and S2 are two stacks with tops are TOP1 and TOP2 respectively. Then write an algorithm to create a new stack S by popping elements from S1 and S2. 5
- Q.3. (a) What is polish notation? What are the advantages of using it? 2
(b) Write an algorithm that converts an infix expression to its prefix equivalent. 5
(c) Suppose a 5-element circular queue with index starts from 0 has front and rear pointers are at 3 and 1 respectively. Then find contents of the queue and the position of front and rear pointers after each of the following statement:
DEQUEUE(); DEQUEUE(); DEQUEUE()
ENQUEUE(10); ENQUEUE(20); ENQUEUE(30); 5
- Q.4. (a) What is recursion? 2
(b) Find 5C_2 using recursion technique. Remember that ${}^nC_1 = n$ and ${}^nC_n = 1$ 4
(c) Define Ackermann's function. Calculate the value of Ackermann function, A(1,3). 6

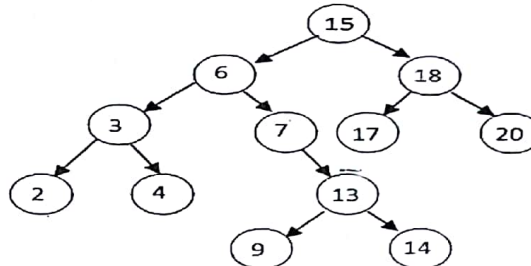
SECTION : B

- Q.5. (a) What is Golden ratio? Give some real world scenario of Golden ratio. 2
(b) Suppose there are two circular link lists headed by h1 and h2 respectively. Then write an algorithm that merges the 2nd list after the 1st list. 4
(c) Write necessary statements to insert a new node pointed by p in a double link list headed by h for the following positions. 6
(i) At the beginning (ii) after a node pointed by q (iii) at the end.
- Q.6. (a) Define for an algorithm (i) time complexity (ii) space complexity 2
(b) Find the complexity of the following function A(n) written in C 4
- ```
void A(n){
 int i,j,k;
 for (i=1;i<=n;i++)
 for(j=1;j<=i2;j++)
 for(k=1;k<=n/2;k++)
 printf("RUET");
}
```
- (c) Write an algorithm to implement Quick-sort technique. Verify the algorithm with the example array a[5]={5,7,6,1,4} 6
- Q.7. (a) What is hashing technique? 2  
(b) Create a hash table using an array with 10 elements for the following numbers implementing quadratic probing technique. 5  
8, 3, 15, 7, 25, 33
- (c) Convert the following binary tree into Maxheap representation. 5



- Q.8. (a) Define heap tree. In what operations, it gives more advantages.  
 (b) For the following BST, briefly describe how to find 9?

3  
4



- (c) Draw the tree representation of the arithmetic expression  $(2 * (a - 1) + (3 * b))$   
 (d) Briefly describe (i) linear data structure and (ii) non-linear data structure

2  
3

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Heaven's Light Is Our Guide  
**RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY**  
**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**  
**1<sup>st</sup> Year Even Semester Examination 2017**

COURSE NO: CSE 1201

COURSE TITLE: Data Structure

FULL MARKS: 72

TIME: 3 HRS

- N.B. (i) Answer any SIX questions taking any THREE from each section.  
(ii) Figures in the right margin indicate full marks.  
(iii) Use separate answer script for each section.

**SECTION : A**

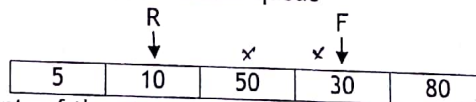
- Q.1. (a) Define data structure and hence describe data structure operations with example. 6  
(b) Explain the following terms: (i) time complexity and (ii) space complexity of an algorithm. 4  
(c) A program p reads in 500 integers in range [0 100] representing the scores of 500 students. It then prints the frequency of each score above 50. What would be the best way for p to store the frequencies? 2  
Q.2. (a) What is the basic difference between traversing and searching? 2  
(b) Suppose you have a unsorted array of size 100. Now you need to search one data from them. Then explain which one of the followings is the best way to do so and why? (i) sort the array and search and (ii) directly search the unsorted array. 4  
(c) What does the following declaration mean? 3

```
int (*ptr)[10];
#include<stdio.h>
int main()
{
 int a[5]={5,1,15,20,25};
 int i,j,m;
 i = ++a[1];
 j = a[1]++;
 m = a[i++];
 printf("%d, %d, %d", i, j, m);
 return 0;
}
```

- Find the output of the following program 3  
(d) Explain the problem of using array data structure in programming and how to solve this problem. 3  
Q.3. (a) Differentiate between list and array. 2  
(b) What is sparse matrix? 1  
(c) Find the triplet representation of the following matrix. 4

$$\begin{matrix} & \begin{matrix} 0 & 0 & 0 & 0 & 0 & 8 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 8 \\ 0 & 9 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

- (d) Write an algorithm using flowchart to find the triplex representation of a sparse matrix. 5  
Q.4. (a) Why Butterfly is so appealing? 2  
(b) Consider the current scenario of a circular queue 5



- Then list the elements of the queue after the execution of the following statement sequentially: Dequeue; Enqueue(18); Enqueue(25); Dequeue();  
(c) Write an algorithm that implements a stack using a link list. 5

**SECTION : B**

- Q.5. (a) With example, explain the followings: (i) best case, (ii) average case and (iii) worst case complexity of an algorithm. 3  
(b) Suppose you have the following pseudocode. Now you find out the estimated time complexity for a particular CPU of 2.5 GHz clock speed and hence make necessary comments. 5

Algorithm: arrayMax(A,n)  
Input: array A of n integers  
Output: maximum element of A

```

currentMax ← A[0]
for i ← 1 to n-1 do
 if A[i] > currentMax then
 currentMax ← A[i]
return currentMax

```

(c) Write the runtime complexity of the following program segments: 4

```

(i) for(i=1; i<n; i=i*2 {...});
(ii) for(i=0; i<n; i++)
 for(j=0; j<n; j++)
 for(k=0; k<n; k++)
 sum++;
(iii) for(int i=0; i<n; i++) {
 n = n/3;
 sum++;
}
(iv) if(condition)
 for(int i = 0; i<n; i++)
 sum++;
Else
 for(int i=0; i<n; i++)
 for(int j=0; j<n; j++)
 sum++;

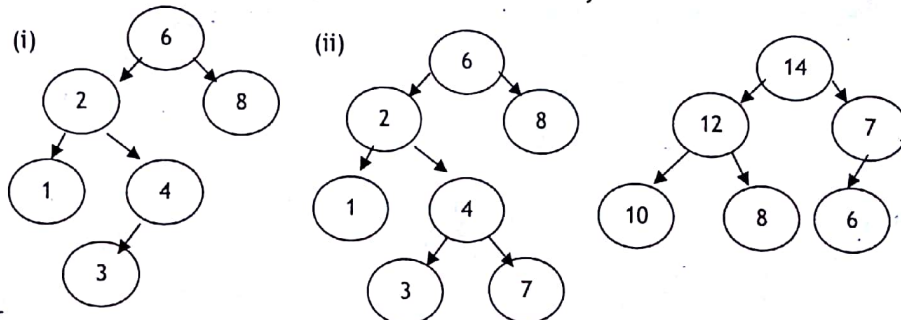
```

- Q.6. (a) For a tree, define (i) Sub-tree, (ii) Traversing and (iii) Levels. 3  
 (b) What is the minimum height of a binary tree with N nodes? 2  
 (c) Draw the BST that is created if the following numbers are inserted in the tree in the given order: 12, 15, 3, 35, 21, 42, 14. Also draw its equivalent balanced BST. 3  
 (d) The following array representation a binary tree with first element as the root. Then sort the array in ascending order using max heap technique. 4

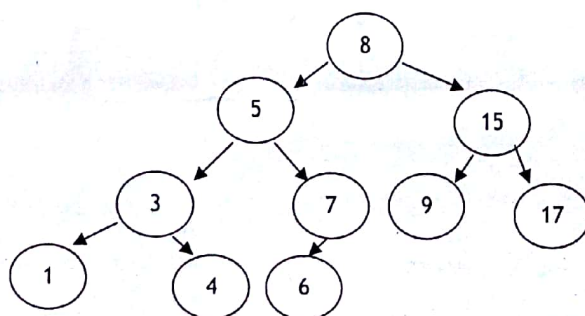
|    |    |    |   |    |    |
|----|----|----|---|----|----|
| 15 | 19 | 10 | 8 | 17 | 16 |
|----|----|----|---|----|----|

- Q.7. (a) What are the properties of a B-tree? 2  
 (b) Write down the steps to sort the following numbers using Quick-Sort technique: 4  
 5, 7, 6, 1, 3, 2, 4  
 (c) Draw a flowchart to implement merge sort for n elements. 4  
 (d) Find the following complexities for merge sort with n elements: (i) Worst-Case time complexity, (ii) Best-Case time complexity, (iii) Average case time complexity and (iv) Space complexity. 2

- Q.8. (a) What is difference between binary tree and binary search tree? Explain with example. 3  
 (b) Write the name of following trees and also write why? 3



- (c) Define with example (i) Max heap and (ii) Min heap for a tree 3  
 (d) For the following complete binary search tree, describe the operations as mentioned below: (i) insert 8 and (ii) delete 6. 3



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